Sang Hak Lee

Marine Data Engineer · Vessel & weather data collection, datasets and quality

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Data engineer who **collects and cleans** AIS / V-PASS vessel data and marine-weather forecast data (NetCDF), and **builds datasets** on PostgreSQL data models I design. Embeds spec-based **data quality validation** and processing **metadata management** into pipelines; licensed weather forecaster with meteorology and ocean-environment certifications, bringing domain understanding of vessel, weather and route data.

EXPERIENCE

GeoSR (Geo System Research) · Data Engineer

Feb 2023 - Present

- **Vessel data collection & cleansing** — Cleaned TB-scale AIS trajectories for the Busan area (outliers, duplicates, time alignment) and joined route/anchorage data to build anchoring, loitering and route-deviation candidate datasets (GeoPandas, multiprocessing).
- **V-PASS dataset construction** — Streaming merge and filtering of 2023-2025 vessel observations into regional/grid statistical datasets and density maps, joined with catch-grade data.
- **Weather & ocean data pipeline** — Automated NetCDF/CSV ingestion → spec validation → raster and current-vector conversion → PostgreSQL loading → GeoServer publishing (Python, Shell, Docker, air-gapped).
- **Data modeling** — Designed PostgreSQL tables and schemas; implemented upserts, transactions and reprocessing.
- **Data quality validation** — Item-level spec reports (variables, dimensions, units), exclusion and count reconciliation, service-point to model-grid matching checks, old vs. new product comparisons.
- **Metadata management** — Maintained processing/error history, input spec sheets and data specifications to keep lineage traceable.
- **Systems & documentation** — Built forecast production servers (Oracle DB, Tomcat), scheduling, monitoring, incident response; requirements and test documents; conference poster (2024), workshop talk (2025).

Korea Meteorological Industry Promotion Agency · R&D Planning Intern

Jun - Sep 2022

- Researched weather technology and policy trends; wrote monthly newsletters and quarterly reports.

SELECTED PROJECTS

Ocean Forecast GeoPipeline

2025.06 - 2026.04

Automated spec checks, spatial conversion, DB loading and layer publishing for NetCDF forecast products, with processing history.

AIS Trajectory Analytics

2024 - 2025

Activity-candidate detection combining AIS tracks with route and anchorage polygons; poster at the KOSMEE autumn conference.

V-PASS Spatial Analytics

2023 - 2025

Turned vessel observations into regional/grid statistics and density maps, with public validation records.

SKILLS

Data	Python, SQL, PostgreSQL, Oracle, Pandas, GeoPandas
Formats	NetCDF, AIS, V-PASS, GeoTIFF, GeoServer, QGIS
Infra	Linux, Shell, Docker, scheduling & monitoring
Other	Fortran, Rust, LLM-based AI agent development

CREDENTIALS · EDUCATION

Weather Forecaster License · Engineer Meteorology · Engineer Ocean Environment

B.S. Earth & Environmental Sciences, Jeonbuk National University (2016 - 2022)

Research Club Grand Prize, JBNU College of Natural Sciences (2021)